Design Verification Plan and Report — T4R1FLASH (virtual test)

Case: `t4_r1_flash` | Generation date: 2026-08-25 | Doc: VBF-T4R1FLASH-DVPR-01

All values mechanical from simulation outputs / parameter set (see log.jsonl); no numbers from memory.

#	Verification item	Condition	Result	Determination	Source
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1	1C discharge capacity	1C CC discharge 25 degC to 2.5 V	3.9547 Ah	PASS (≥ nominal 3.93 Ah)	cell/final_b7_1c_dfn.json
2	-20 degC capacity retention	1C discharge at 253.15 K / 1C at 298.15 K	0.99286	PASS (≥ 0.95)	cell/final_b7_retention_dfn.json
3	Gravimetric energy density	∫V·I dt / contract mass	460.06 Wh/kg	PASS (≥ 327.18)	cell/final_b7_energy_dfn.json
4	Volumetric energy density	∫V·I dt / layer-stack volume	899.25 Wh/L	PASS (≥ 880.0)	cell/final_b7_energy_dfn.json
5	4C fast-charge temperature rise	4C CC charge, 45 degC ambient, lumped thermal	T_max 329.61 K	PASS (≤ 333.15 K)	cell/final_b7_4c45_dfn.json
6	4C fast-charge plating	anode potential min over 4C charge	+0.0193 V	PASS (> 0 V, plated=false)	cell/final_b7_4c45_dfn.json
7	Overcharge -> thermal runaway	0.5C overcharge to 4.7 V + TR ODE	triggered = False	PASS (no runaway)	cell/final_b7_overcharge.json, cell/final_b7_tr.json
8	Voltage window	parameter set cut-offs	2.5–4.2 V	PASS	parameter set
9	Nail penetration	—	N/A (beyond pure simulation boundary, requires physical experiment)	—	—
10	Crush / drop	—	N/A (beyond pure simulation boundary, requires physical experiment)	—	—
11	Cycle life	—	N/A (no aging model in this case)	—	—
12	Rate-pulse internal resistance	—	N/A (beyond pure simulation boundary)	—	—

Conclusion

8/8 simulation-verifiable items PASS at DFN precision (model_used=DFN, no fallback). Uncovered items: nail penetration, crush, drop, cycle life, rate-pulse DCR — physical-experiment boundary (cited as paper limitations).